Eberhard Karls Universität Tübingen | Seminar für Sprachwissenschaft (SfS)
Methods I: Logic, WS 2026/27 | Dr. Kyle Richardson
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Exercise Sheet 1
Naïve Set Theory

Exercise 1: [5 points] At the University of Tübingen, 60 students were asked whether they were taking German or Spanish language courses. The results were as follows:

- **35 students** are taking German.
- **28 students** are taking Spanish.
- **12 students** are taking *both* courses.

Let G be the set of students taking German and S the set of students taking Spanish.

Given these details, answer the following questions:

(a) Draw a Venn diagram to illustrate the situation, including students taking neither course.



(b) How many students are taking only German?

(b) _____

(c) How many students are taking exactly one of the two courses?

(c) _____

(d) How many students are taking neither German nor Spanish?

(d) _____

(e) Express the set of students taking German but not Spanish using set notation.

(e) _____

Exercise 2: [6 points] Let $A = \{\varnothing, \{\varnothing\}, a\}$ and $B = \{\{\varnothing\}, \{a\}, a\}$.
Answer the following:

(a) i. Is $\{\varnothing\} \in A$?

i. _____

ii. Is $\{\varnothing\} \subseteq A$?

ii. _____

iii. Justify your answer for parts i and ii.

_____

_____

(b) Write out the complete power set $\mathcal{P}(A)$.

(c) i. Find $A \cap B$.

i. _____

ii. Find $A \setminus B$.

ii. _____

iii. Find $B \setminus A$.

iii. _____

iv. What is the cardinality of $\mathcal{P}(A \cap B)$?

iv. _____

(d) i. Is $A \in \mathcal{P}(A)$?

i. _____

ii. Is $A \subseteq \mathcal{P}(A)$?

ii. _____

iii. Is $\{\varnothing\} \in \mathcal{P}(A)$?

iii. _____

iv. Justify your answer for $A \subseteq \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$ part.

Exercise 3: [3 points] Define the following sets using the three main methods of defining sets (*listing elements, characteristic property, and recursive definition*).

Each set must be defined using one, best-fitting method, and you should not use any method more than once.

(a) D is the set of all positive integers divisible by 4.

(b) $E = \{x \in \mathbb{Z} \mid -4 \leq x \leq 4 \text{ and } x \text{ is even}\}$. [1]

[1] Define E using a different method than it is currently defined with.

(c) F is the set of all rational numbers strictly between 0 and 1.

Exercise 4: [4 points] Let A, B, C be arbitrary, non-empty sets. Let $\mathcal{U}$ be $A \cup B \cup C$.
Say whether the following are *true* or *false*:

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(a) _____

(b) $(A \cap B) \setminus C = A \cap (B \setminus C)$

(b) _____

(c) $A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)$

(c) _____

(d) $A \cap (B \setminus A) = \varnothing$

(d) _____

Exercise 5: [2 points] Decide whether the following are well-formed sets or not (write *yes* or *no*):

(a) $\{a, b, \{c, d\}\}$

(a) _____

(b) $\{a, b, c$

(b) _____

(c) $\{\varnothing, \{\varnothing\}, a\}$

(c) _____

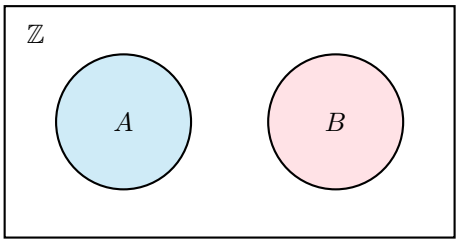
(d) $\langle a, b, c\rangle$

(d) _____

Exercise 6: [3 points] Consider the following sets of integers:

$$A = \{x \in \mathbb{Z} \mid x \text{ is even}\}, \qquad B = \{x \in \mathbb{Z} \mid x \text{ is a multiple of } 3\}.$$

The Venn diagram below is **incorrect**.



(a) Draw the **correct Venn diagram** representing A and B.

(b) Give one integer that demonstrates why the original diagram is incorrect, and briefly explain your answer.